

02_DATA_FLOW_AND_CLOSED_LOOP

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EVIDENCE PROVENANCE: [CLOSED-LOOP CAUSALITY]

Complete mapping of environmental fog guidance, V2V peer-to-peer telemetry, and V2I supervisory data flows.

1. The Three Primary Information Loops

- Loop 1: Fog-to-Driver Action:** Weather Station measures optical visibility -> Ingestion Gateway -> Digital Twin -> Safety Solver calculates v_{safe} -> Cockpit Display alerts operator -> Driver applies brake.
- Loop 2: Dual-Vehicle V2V Loop:** Truck 01 (500ms slot) and Truck 02 (1000ms slot) exchange 11-field kinematic packets over 433 MHz LoRa to maintain peer-to-peer headway awareness.
- Loop 3: V2I Supervisory Loop:** Vehicles transmit 10 Hz telemetry over Wi-Fi REST to the central twin, which streams live 3D positions to the Control Room console via WebSocket.

2. Timing & Latency Budget

Subsystem Stage	Latency Allowance	Measured Empirical Latency
Sensor Ingestion & Debounce	15.0 ms	10.0 ms
ESP32 Wi-Fi Transmission	25.0 ms	12.4 ms
Backend Physics & Solver	10.0 ms	5.5 ms
WebSocket State Broadcast	20.0 ms	14.5 ms
Frontend Render & Display	50.0 ms	40.0 ms
TOTAL END-TO-END LATENCY	< 250.0 ms	82.4 ms (VERIFIED)